

Microeconomics Through the Lens of EV and CV

A Coherent Map with Sufficient Detail

Big Picture: Why EV/CV Can Be a “Spine”

EV (equivalent variation) and CV (compensating variation) are *money-metric welfare* measures. They force you to build the full consumer-theory machinery:

preferences $\Rightarrow$ **optimization** $\Rightarrow$ **dual value functions** $\Rightarrow$ **derivative identities**
 $\Rightarrow$ **mechanisms (income vs substitution)** $\Rightarrow$ **welfare in dollars** $\Rightarrow$ **testability (revealed preference)**.

The key organizing principle is **duality**: switching between “choose goods given money” and “choose money given utility”.

Throughout, let $p \in \mathbb{R}_{++}^n$ be prices, $m > 0$ wealth, $x \in \mathbb{R}_+^n$ consumption, and $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ utility representing preferences.

1. The Primitives: Preferences and Feasible Choice

1.1 Utility representation and monotone transformations

Utility is *ordinal*: if u represents preferences, then any strictly increasing $f(u)$ represents the same preferences. This matters because EV/CV are defined via feasibility sets $u(x) \geq \bar{u}$, which are invariant to monotone transforms.

1.2 Budget set

$$B(p, m) = \{x \in \mathbb{R}_+^n : p \cdot x \leq m\}.$$

Under local nonsatiation (“more is better” locally), optimal choices satisfy **budget exhaustion**:

$$p \cdot x^*(p, m) = m.$$

This is the consumer-side “Walras/budget binding” logic.

2. The Main Problem (UMP): Demand and Indirect Utility

2.1 Utility maximization problem (UMP)

$$v(p, m) \equiv \max_{x \geq 0} \{u(x) : p \cdot x \leq m\}.$$

- The optimizer(s) define **Marshallian demand** $x(p, m)$.
- The optimal value defines **indirect utility** $v(p, m)$.

2.2 Basic properties (used later in welfare and aggregation)

- **Homogeneity degree 0:** $x(\alpha p, \alpha m) = x(p, m)$, and $v(\alpha p, \alpha m) = v(p, m)$.
- **Monotonicity:** $v(p, m)$ increases in m and (weakly) decreases in each p_i .

These properties are quick correctness checks for solved demands and ensure money-metric concepts behave sensibly.

2.3 KKT as the universal solver (why it enters the EV/CV spine)

UMP is a constrained optimization problem with inequalities (budget, nonnegativity). KKT provides:

- stationarity (FOCs),
- feasibility,
- complementary slackness (which tells you **interior vs corner**).

EV/CV ultimately depend on the ability to compute $x(p, m)$ and $v(p, m)$, which in general requires KKT/Lagrange methods.

3. The Dual Problem (EMP): Hicksian Demand and the Expenditure Function

3.1 Expenditure minimization problem (EMP)

Fix a target utility $\bar{u}$.

$$e(p, \bar{u}) \equiv \min_{x \geq 0} \{p \cdot x : u(x) \geq \bar{u}\}.$$

- Minimizers define **Hicksian (compensated) demand** $h(p, \bar{u})$.
- The value defines the **expenditure function** $e(p, \bar{u})$.

3.2 Why EMP is essential for EV/CV

EV/CV are defined as the *money changes* needed to hit specific utility targets at specific prices. That is exactly what $e(p, \bar{u})$ computes.

3.3 Properties that make EV/CV well-behaved

- **Homogeneity degree 1 in prices:** $e(\alpha p, \bar{u}) = \alpha e(p, \bar{u})$.
- **Monotonicity:** $e(p, \bar{u})$ increases in $\bar{u}$ and is nondecreasing in each p_i .
- **Concavity in prices:** $e(\cdot, \bar{u})$ is concave in p (drives Slutsky symmetry/NSD).
- **Target achieved:** at the optimum $u(h(p, \bar{u})) = \bar{u}$ (under monotonicity).

4. Duality: The Bridge that Makes EV/CV Possible

Duality gives a *dictionary* between the UMP world (x, v) and the EMP world (h, e) .

4.1 Core identities

$$\begin{aligned} e(p, v(p, m)) &= m, & v(p, e(p, \bar{u})) &= \bar{u}, \\ x(p, m) &= h(p, v(p, m)), & h(p, \bar{u}) &= x(p, e(p, \bar{u})). \end{aligned}$$

4.2 Interpretation

- “Start with money m , get max utility v ; then ask the minimum money to reach that utility at the same prices” must return m .
- “Start with target utility $\bar{u}$, compute minimum spending e ; then maximize utility with that spending” must return $\bar{u}$.

EV/CV are built by comparing expenditures needed to reach u_0 and u_1 under old/new prices.

5. Envelope Logic: Turning Value Functions into Behavior

EV/CV are welfare objects, but they are computed and interpreted using derivative identities.

5.1 Shephard’s lemma (EMP)

If e is differentiable,

$$\frac{\partial e(p, \bar{u})}{\partial p_i} = h_i(p, \bar{u}).$$

Meaning: the marginal cost of raising price p_i equals the compensated quantity of good i used at the cost-minimizing bundle.

5.2 Roy’s identity (UMP)

If v is differentiable,

$$x_i(p, m) = - \frac{\partial v(p, m) / \partial p_i}{\partial v(p, m) / \partial m}.$$

Meaning: Marshallian demands can be recovered from how the maximal attainable utility changes with prices and income.

6. Mechanisms: Substitution vs Income (Slutsky) and Their Link to EV/CV

6.1 Slutsky decomposition

Let $\bar{u} = v(p, m)$. Then:

$$\frac{\partial x_i(p, m)}{\partial p_j} = \underbrace{\frac{\partial h_i(p, \bar{u})}{\partial p_j}}_{\text{substitution effect}} - \underbrace{\frac{\partial x_i(p, m)}{\partial m} x_j(p, m)}_{\text{income effect}}.$$

- **Substitution effect:** response holding utility fixed (Hicksian world).
- **Income effect:** extra response because the price change alters purchasing power.

6.2 Slutsky matrix and curvature

Define

$$S_{ij}(p, m) = \frac{\partial x_i}{\partial p_j} + \frac{\partial x_i}{\partial m} x_j.$$

Then

$$S_{ij}(p, m) = \frac{\partial h_i(p, v(p, m))}{\partial p_j} = \frac{\partial^2 e(p, v(p, m))}{\partial p_i \partial p_j}.$$

Because $e(\cdot, \bar{u})$ is concave in prices, S is **symmetric** and **negative semidefinite**.

6.3 Direct connection: EV/CV and income effects

- If **income effects are zero** (e.g., quasi-linear preferences), then Hicksian and Marshallian responses coincide and typically

$$EV = CV.$$

- When income effects matter, EV and CV generally differ because they evaluate the welfare change using different price systems and different utility reference levels.

7. EV and CV: Definitions, Interpretations, and Integral Formulas

7.1 Define baseline and post-change utilities

Prices change from p to $\tilde{p}$, wealth remains m :

$$u_0 \equiv v(p, m), \quad u_1 \equiv v(\tilde{p}, m).$$

Typically under a price increase, $u_1 < u_0$.

7.2 Compensating variation (CV)

Definition: money at new prices needed to restore baseline utility:

$$v(\tilde{p}, m + CV) = u_0.$$

By duality:

$$m + CV = e(\tilde{p}, u_0) \quad \Rightarrow \quad \boxed{CV = e(\tilde{p}, u_0) - m.}$$

Interpretation: “After the price change, how much compensation is required to make the consumer as well-off as before?”

7.3 Equivalent variation (EV)

Definition: money at old prices that is equivalent to the price change:

$$v(p, m - EV) = u_1.$$

By duality:

$$m - EV = e(p, u_1) \quad \Rightarrow \quad \boxed{EV = m - e(p, u_1) = e(p, u_0) - e(p, u_1).}$$

Interpretation: “Before the price change, how much money would you take away to make the consumer as well-off as after the change?”

7.4 Integral (Hicksian) representation for a single price change

If only good 1's price changes from p_1 to $\tilde{p}_1$ (other prices p_{-1} fixed), then:

$$EV = \int_{p_1}^{\tilde{p}_1} h_1(t, p_{-1}, u_1) dt, \quad CV = \int_{p_1}^{\tilde{p}_1} h_1(t, p_{-1}, u_0) dt.$$

Key message: EV and CV integrate *compensated demand*, but at *different utility levels*.

7.5 Comparison and special cases

- Small price changes: EV and CV are typically very close (difference is second-order).
- Quasi-linear (no income effects): $EV = CV$ and both coincide with “exact” consumer surplus.
- With significant income effects: EV and CV differ; the gap is informative about the role of income effects and curvature.

8. From Welfare to Data: Revealed Preference as “Testability”

8.1 Why revealed preference belongs on the same spine

EV/CV assume choices come from optimization. Revealed preference asks whether observed choices could have come from *some* $u(\cdot)$.

8.2 WARP/SARP

- **WARP:** rules out pairwise contradictions in observed choices (no two-budget strict reversals).
- **SARP:** rules out cycles across multiple observations; stronger consistency requirement.

Connection: if data violate these, then the UMP/EMP framework (and hence EV/CV) cannot rationalize the choices.

9. Aggregation: When Welfare and Demand “Scale Up”

9.1 Aggregate demand

With I consumers:

$$X(p; m_1, \dots, m_I) = \sum_{i=1}^I x_i(p, m_i).$$

In general, aggregation depends on the *distribution* of wealth, not only total wealth.

9.2 Gorman form and aggregation

If

$$v_i(p, m_i) = a_i(p) + b(p)m_i$$

with the same $b(p)$ across consumers, then aggregate demand depends only on $\sum_i m_i$. **Connection:** “income effects” must align across consumers for clean aggregation.

10. Where This Spine Stops (and What It Mirrors)

10.1 Production side analogy

On the firm side, the analogous objects are profit and cost functions:

$$\pi(p) = \max_{y \in Y} p \cdot y, \quad C(w, q) = \min_{z \geq 0} \{w \cdot z : f(z) \geq q\},$$

with envelope-type results (Hotelling/Shephard). This mirrors the consumer duality, but EV/CV themselves are consumer welfare measures.

10.2 General equilibrium

Competitive equilibrium combines:

- consumer optimality (UMP),
- firm optimality (PMP),
- market clearing.

Welfare theorems link equilibrium and Pareto efficiency. EV/CV are then used to *quantify* welfare changes under policy or price shifts in equilibrium environments.

One-Page Mental Checklist (EV/CV-Centered)

1. Identify the policy/price change: $p \rightarrow \tilde{p}$ and baseline wealth m .
2. Compute $u_0 = v(p, m)$ and $u_1 = v(\tilde{p}, m)$ (consumer welfare before/after).
3. Translate welfare to money via $e(\cdot)$:

$$EV = m - e(p, u_1), \quad CV = e(\tilde{p}, u_0) - m.$$

4. If needed, compute via compensated demand integrals:

$$EV = \int h_1(\cdot, u_1), \quad CV = \int h_1(\cdot, u_0).$$

5. Interpret differences using income effects:
 - no income effects $\Rightarrow EV=CV$,
 - strong income effects $\Rightarrow EV$ and CV can differ materially.
6. Check consistency with data (WARP/SARP) if the analysis is empirical.